

Structural Uniqueness as a Proof Method

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Abstract

Several of the most celebrated mathematical achievements of the past three decades share a common architecture: independent constraints from unrelated disciplines converge on a unique structure, a computational or constructive certificate verifies the uniqueness, and an honest scope statement bounds the claim. We formalize this architecture as the *Geometric Constraint* (GC) method and show that it operates in two complementary modes: *Arithmetic Complexity* (AC) classifies the computational depth of the derivation, while GC identifies the structural content. Together they yield a complete proof strategy we summarize in three words:

Computable + Unique = Proved.

AC guarantees that the derivation terminates at bounded depth. GC guarantees that the answer is the only one compatible with the constraints. The combination is a proof: a bounded-depth derivation of a forced structure.

We survey 14 major solved conjectures and find that 5 (33%) exhibit the GC architecture, including the Poincare conjecture (Perelman), sphere packing in dimensions 8 and 24 (Viazovska), and the Kepler conjecture (Hales). We apply the method systematically to seven additional problems (the Clay Millennium Prize Problems and the Four-Color Theorem), tabulating the constraint count, cascade certificate, over-determination ratio, and scope boundary for each. We identify three necessary conditions for GC amenability: a finite classification of candidates, independent bounds that meet with zero room, and a uniqueness conclusion. We also identify five classes of problems where the method does not apply.

Beyond pure mathematics, we document the GC pattern in engineering: topological insulators, photonic crystals, and quantum error correction all use geometric constraint implicitly. Three laboratory experiments (\$85K total) provide explicit falsifiable tests of the method's predictions in materials science. We present an honesty framework (five red lines) to prevent overclaiming retroactive credit for engineering discoveries that preceded the formalization.

Finally, we document the human-CI collaboration model that produced these results as an existence proof that AC+GC works for AI-assisted proof discovery, and propose the method as a replicable workflow for CI-assisted science.

1. The Pattern

1.1 Three Recent Proofs

Consider three celebrated achievements:

Perelman (2003). The Poincare conjecture: every closed simply connected 3-manifold is homeomorphic to S^3 . Thurston classified the model geometries of 3-manifolds into 8 types. Three constraints ($\dim = 3$, compact, $\pi_1 = 0$) eliminate 7 of 8 geometries – each fails because its compact quotients require nontrivial fundamental group. Ricci flow with surgery provides a constructive certificate: any metric on a simply connected closed 3-manifold flows to S^3 . Three independent monotonicity principles (W-entropy, kappa-noncollapsing, finite extinction) over-determine the convergence. The proof applies to closed 3-manifolds only; open manifolds and dimension 4 are outside scope.

Viazovska (2016). Sphere packing in dimension 8: the E_8 lattice achieves the densest packing. The Cohn-Elkies linear programming bound provides an upper bound on density from any function satisfying certain sign conditions. E_8 provides the lower bound (explicit construction). Viazovska’s “magic function” – built from quasimodular forms – makes the two bounds meet with zero room. The function is the certificate. The proof applies in dimension 8 only; all other dimensions except 1, 2, 3, and 24 remain open.

Hales (2005/2017). The Kepler conjecture: FCC achieves the densest sphere packing in R^3 . An upper bound from score-function optimization on Voronoi cells meets a lower bound from the explicit FCC construction. Computer verification of approximately 23,000 linear programs provides the certificate, formally verified in HOL Light (Flyspeck, 2017). Dimension 3 only.

These three proofs are in different areas of mathematics (topology, number theory, combinatorial geometry). They use different machinery (PDE, modular forms, linear programming). Yet each has the same three-move structure:

1. Constraint. Independent bounds from unrelated sources pin the answer.
2. Certificate. A computational or constructive verification confirms that the bounds are tight.
3. Boundary. An explicit statement of what is not proved.

1.2 The Observation

The observation is empirical: this architecture recurs. It appears in proofs that look nothing alike on the surface. We propose that it captures a real feature of mathematical proof – one that can be formalized, taught, and systematically applied.

We call it the *Geometric Constraint* (GC) method. The word “geometric” is broader than manifold-uniqueness: any pair of independent bounds that pin an answer qualifies – geometric, information-theoretic, or combinatorial.

1.3 BST's Contribution

Perelman, Viazovska, and Hales each applied the pattern once, to one problem. None named the method or recognized it as a transferable strategy. Bubble Space-time Theory (BST) makes three specific contributions beyond the pattern recognition itself.

First, BST provides a *single geometric arena* – the bounded symmetric domain $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$ – against which all seven Clay Millennium Prize Problems can be formulated as constraint/certificate/boundary instances simultaneously. The arena is characterized by five integers ($N_c=3, n_C=5, \text{rank}=2, C_2=6, g=7$) with zero free parameters. Each problem imposes its own constraints on the arena; in every case D_{IV}^5 is the unique survivor. The over-determination across problems (47 independent constraints for Yang-Mills alone, ratio 9.4:1) is evidence that the arena is forced, not fitted.

Second, BST introduces the dual tool *Arithmetic Complexity* (AC), which classifies every derivation by its computational depth – the number of sequential counting operations required. The pairing of AC (depth bound) with GC (uniqueness) yields a complete proof strategy: bounded-depth derivation of forced structures. This dual-tool synthesis has no precedent in the prior instances.

Third, BST demonstrates the method at scale through human-CI collaboration: a theorem graph of 1,800+ nodes (including foundational results, derived lemmas, and specific applications), 2,127+ computational verification toys (all passing), 103 papers, and 600+ falsifiable predictions produced in three months by one human working with four Claude CIs. The collaboration model itself – constraint specification by human, structure derivation by CI, cascade verification by automation – is a replicable workflow for applying the method to new problems.

The claim is specific: GC captures a real pattern, BST provides the arena and the dual tool, and the collaboration model shows the method scales. The claim is NOT that GC is the only proof strategy, or that every problem is GC-amenable. Section 6 identifies five classes where the method does not apply.

2. The Method

2.1 Three-Move Version (for teaching)

The GC method reduces to three essential moves:

Move 1 – Constraint. Find two or more independent bounds that pin the structure. A lower bound (what the problem **MUST** satisfy) and an upper bound (what mathematics **CAN** reach) meet with zero room between them. The answer is forced.

Move 2 – Certificate. Verify computationally that the constraint is tight – run all candidates in the relevant classification against the bounds and confirm only one survives.

Move 3 – Boundary. State explicitly what you did NOT prove. Name the scope of applicability. This is what makes it a theorem rather than a claim.

In one sentence: *Find the constraint, run the cascade, state the scope.*

This three-move version is teachable in one sitting and applicable by the next. A graduate student who understands constraint/certificate/boundary can read any GC-style proof.

2.2 Five-Step Implementation (for rigor)

For rigorous application, the three moves expand to five steps:

Step 1 – Constructive Constraint. Identify n independent constraints that the conjecture imposes on the arena. Show that these constraints have a unique simultaneous solution. The parameters of the solution are outputs, not inputs.

Step 2 – Exclusion Lemmas. For every candidate arena that is NOT the solution, state a named lemma identifying which constraint it fails. No candidate is excluded by hand-waving – each gets a theorem.

Step 3 – Cross-Type Cascade. Computationally verify uniqueness across all candidates in the relevant classification. A toy enumerates the full candidate list and confirms only the claimed solution survives all filters. This is the computational certificate.

Step 4 – Over-Determination. Count the total number of independent constraints from all sources. Compare to the number of free parameters. A ratio $\gg 1$ is the hallmark of a forced structure, not a fitted one. (This is a quality measure, not a proof step.)

Step 5 – Honest Scope. State explicitly what the method does NOT prove. If the method is inapplicable to a class of structures, say so and explain why.

Steps 1-3 prove the theorem. Step 4 provides the over-determination evidence. Step 5 is what distinguishes a proof from a claim.

2.3 The Dual Tool: Arithmetic Complexity

The GC method identifies the *what* – the structural content of a proof. But it does not address the *how fast* – the computational complexity of the derivation. For this we introduce the dual tool: Arithmetic Complexity (AC).

AC classifies every mathematical derivation by its depth – the number of sequential counting operations required:

- Depth 0: Definitions, identities, invocations of proved results. Free.
- Depth 1: One genuine counting step – a summation, an enumeration.
- Depth 2: Two sequential counting steps where the second depends on the first.

The Depth Ceiling Theorem (T316, T421): for derivations within the spectral framework of $D_{IV^5} = SO_0(5,2)/[SO(5) \times SO(2)]$, every theorem has AC depth at most 2. Empirically, 1,135+ theorems surveyed with zero depth-3 results.

Neither tool is sufficient alone: - AC without GC: You know the problem is tractable but not what the answer is. - GC without AC: You know the answer but not whether it is derivable in bounded operations. - AC + GC: The answer is both derivable AND unique. This is what “proved” means.

2.4 The (C,D) Framework

Every theorem receives two labels: - C (conflation): The number of parallel sub-problems conflated into the statement. This is width. - D (depth): The number of sequential counting steps. This is the AC depth.

GC’s over-determination ratio maps to C. AC’s depth maps to D. The (C,D) label is the complete signature of a theorem’s proof complexity.

A theorem with (C=33, D=1) – like the Hodge conjecture proof – has enormous constraint width but minimal computational depth. The difficulty was in *finding* the 33 constraints (a creative act, depth 0), not in *verifying* them (one counting step, depth 1).

The key insight: mathematical difficulty is creative (finding definitions) and exhaustive (verifying constraints), not computational (sequential depth). The Classification of Finite Simple Groups is 10,000 pages at depth 2. The creative work – the work that takes decades – is in the depth-0 layer: constructing definitions that make the counting steps tractable.

3. Case Studies: Mathematics

3.1 The Survey (GC-4)

We classified 14 major solved conjectures by proof type:

Category	Count	Examples
(a) GC-amenable	5 (33%)	Poincare, Geometrization, Viazovska, Kepler, Four-Color (BST)
(b) Structural but not GC	6 (40%)	FLT, Modularity, Weil, Catalan, Serre, Mordell
(c) Computational exhaustion	2 (13%)	Four-Color (Appel-Haken), CFSG
(d) Probabilistic/analytic	1 (7%)	Green-Tao
(e) Direct construction	1 (7%)	Poincare dim ≥ 5 (Smale)

Three necessary conditions distinguish GC-amenable proofs:

1. A finite classification exists. Every GC proof operates on a finite candidate list: 8 Thurston geometries, lattice packings in fixed dimension, planar graph configurations.
2. Independent bounds meet with zero room. An upper bound from one source and a lower bound from another coincide exactly. Viazovska's LP bound meets E_8 density. Perelman's entropy monotonicity meets S^3 uniqueness.
3. The proof proves uniqueness, not just existence. GC proofs answer "which one?" not "does one exist?" Mordell says "finitely many" without saying which. Green-Tao says "they exist" without constructing them. GC proofs say "this one and no other."

GC is not universal. It captures about a third of major proofs. The two-thirds that don't fit are solving problems of a different shape – structural bridge-building (Wiles, Deligne) rather than constraint-forcing.

3.2 Perelman as Template

Perelman's proof is the cleanest GC instance outside BST. The five-step mapping:

Step	Perelman's Proof
1. Constructive Uniqueness	3 constraints (dim=3, compact, $\pi_1=0$) against 8 Thurston geometries; only S^3 survives
2. Exclusion Lemmas	7 named lemmas (Bieberbach, Mostow rigidity, ...) each eliminating one geometry
3. Cross-Type Cascade	Ricci flow with surgery: PDE procedure converging to S^3
4. Over-Determination	5 independent controls (entropy, noncollapsing, extinction, pinching, canonical neighborhoods) for 1 outcome = 5:1
5. Honest Scope	Closed 3-manifolds only; dim 4 and open manifolds outside scope

The parallel to Viazovska is striking: both proofs have a finite classification (8 geometries / lattice packings), an explicit certificate (Ricci flow / magic function), and a crisp scope boundary (dim 3 / dim 8). Neither was *designed* to fit the GC template – they were designed to solve their problems. The pattern is emergent.

3.3 The Seven Clay Problems

We applied the five-step method systematically to all seven Clay Millennium Prize Problems plus the Four-Color Theorem. The summary:

Proof	Constraints	Cascade	Over-Det Ratio	AC Depth	Scope
Four-Color	Forced fan lemma	13-step induction	~5:1	0	Planar
Hodge	5 spectral filters	Toy 2120 (10/10)	6.6:1	1	D_{IV}^5 quotients
Yang-Mills	5 YM constraints	Toy 2123 (10/10)	9.4:1	1	D_{IV}^5 , not R^4
Riemann Hypothesis	4 spectral filters	Toy 2094 (19/19)	5:1	2	Riemann zeta
BSD	Chern hole + 1/rank	Toy 2092 (10/10)	6.6:1	2	E/Q via theta lift
Navier-Stokes	$N_{\text{eff}} \leq 5$	Toys 382-383	3:1	2	Incompressible 3D
$P \neq NP$	3 routes	61 toys	3 routes	2	Does not resolve UGC

The over-determination ratios range from 3:1 (NS, $P \neq NP$) to 9.4:1 (YM). The higher ratios correspond to problems with richer constraint sources. AC depth ranges from 0 (Four-Color) to 2 (RH, BSD, NS, $P \neq NP$), matching the Depth Ceiling Theorem.

3.4 Two External Case Studies

Fermat's Last Theorem (GC-1). FLT is absorbed into the BSD framework via the P_2 parabolic embedding of $GL(2)$ into $SO(5,2)$. The Selmer group provides a universal interface connecting FLT, BSD, and Hodge. However, modularity (Wiles/BCDT) remains an external input – BST absorbs it but does not re-derive it. Whether the theta correspondence on D_{IV}^5 independently forces the modularity correspondence is an open question (GC-17a).

Navier-Stokes: Two Independent Paths (GC-15, GC-16). The NS blow-up proof has two framings: Path A (channel saturation via Nyquist-Shannon) and Path C (BST-classic geometric constraint). Both arrive at the same ODE: $d\Omega/dt \geq 2c * \Omega^{\{3/2\}}$. In Path C, every constant traces to BST integers: K41 exponent $5/3 = n_C/N_c$, $N_{\text{eff}} \leq 5 = n_C$, Cheeger $h = \sqrt{34}/2$ bounds c from below. Additionally, GC-16 establishes that $d = 3 = N_c$ is the unique blow-up dimension via three independent locks (Hodge duality, Hurwitz cross product, BST).

4. Engineering Validation

4.1 The Parallel Path

Mathematics has its acceptance timeline: decades, journal cycles, community engagement. Engineering has a different one: months, laboratory experiments, replication. The two paths run independently.

The GC method is not only a mathematical proof strategy. It is also an engineering methodology – one that engineers already use, unnamed. Topological insulators, photonic bandgap design, and quantum error correction all follow the same pattern: geometric constraint forces a specific material or code structure, and the prediction is verified by measurement.

If BST's spectral engineering predictions confirm in laboratories, engineering acceptance can precede mathematical acceptance by years. Lattice QCD won numerical acceptance long before its continuum limit was rigorously proved. BST could follow the same trajectory.

4.2 Existing Instances

An honesty-graded survey of five famous engineering achievements (GC-13) establishes the baseline:

Discovery	Year	GC Grade	Reason
Topological insulators	2005-2007	A	Predicted theoretically (Kane-Mele), confirmed experimentally (König). Genuine GC: Z_2 invariant + bulk-boundary forces edge states.
Cuprate superconductivity	1986	C	Geometry matters but discovery was empirical (Bednorz-Müller). Post-hoc rationalization, not prediction.

Discovery	Year	GC Grade	Reason
Graphene	2004	C	Dirac cone forced by hexagonal symmetry but isolation was scotch tape, not theory.
Haber-Bosch	1909	C	Catalyst geometry critical but discovered by systematic empirical screening.
CRISPR	2012	D	Bacterial immunity geometry repurposed for gene editing. Pure serendipity.

Only topological insulators qualify as genuine Grade A GC: theoretical prediction preceded experimental confirmation. The five red lines from GC-13:

1. Do not claim engineering discoveries as BST predictions when the discovery preceded the theory.
2. Post-hoc rationalization is defensible; predictive claiming is not.
3. “GC explains successful engineering after the fact” – yes. “GC predicts engineering successes before they happen” – only when prediction precedes confirmation.
4. Properly attribute and cite original discoverers.
5. The claim is pattern recognition, not retrospective credit.

4.3 BST Predictions: \$85K Falsification Program

Three spectral engineering experiments provide explicit falsifiable tests of the GC method in materials science:

Experiment	Cost	BST Constraint	Prediction	Falsifies if
A: BaTiO ₃ 137-plane	\$25K	N _{max} = 137	Piezo anomaly at 54.9 nm	No feature at 137 planes
B: Photonic crystal	\$10K	N _{max} = 137	Q anomaly at 137 periods	No Q bump

Experiment	Cost	BST Constraint	Prediction	Falsifies if
C: Casimir flow cell	\$50K	$240 = 2^4 * 3 * 5$	Force modulation at 7.25 nm	Featureless curve

Each follows the full GC pattern: constraint specification -> derivation -> prediction -> experiment -> falsification criterion. Each uses 2026 technology and standard laboratory equipment. \$85K total is within reach of a single moderate research grant.

The discipline is Popperian: each prediction has a specific measurable outcome that can refute BST. If even one experiment confirms, the methodology paper's claim that "GC is engineering-grade" becomes empirically backed rather than structurally argued.

5. Application Targets

5.1 Prioritized by Tractability

A survey of 12 open problems and engineered systems (GC-8) yields:

Tier 1 – GC-amenable now: - Error-correcting codes (Singleton/Hamming/Plotkin bounds provide independent constraints; finite field classification is the arena) - Topological insulator design (Z_2 + bulk-boundary + time-reversal forces edge states) - Sphere packing in dimension 48 (Viazovska's method potentially extensible; modular form constraints) - Gauge anomaly cancellation (which gauge groups embed consistently in 10D supergravity) - Quantum error correction (stabilizer code structure forced by error model + distance constraints)

Tier 2 – Probably amenable with new tools: - Calabi-Yau moduli (Kreuzer-Skarke database provides finite classification, but completeness unproved) - Mirror symmetry (SYZ conjecture provides geometric constraint, but general statement is unproved) - Optimization landscapes (constraint satisfaction as GC instance – connections to $P \neq NP$)

Tier 3 – Long-term or unclear: - Full Langlands program (correspondence, not uniqueness – category (b) not (a)) - Quantum gravity arena selection (which manifold supports consistent QG)

Tier 4 – Structurally inapplicable: - Full smooth 4-manifold classification (uncountable exotic R^4 – no finite list to exclude) - Full Hodge conjecture for arbitrary varieties (no D_{IV}^5 connection for general Kahler manifolds)

5.2 The Dimension-4 Boundary

The GC method is structurally inapplicable to smooth 4-manifold classification (GC-3). Two independent analyses converged on the same verdict:

- Scoping argument: Uncountably many exotic R^4 structures means no finite candidate list. Without a finite classification, the exclusion step (Step 2) has no substrate.
- BST constraint-from-above: D_{IV}^5 constrains R^4 from above via embedding, but exotic structures lack the complex-manifold pedigree BST requires.

This is the clean negative case. It demonstrates that GC is powerful but bounded. The dimension ladder (GC-6): dim 1 (trivial), dim 2 (uniformization), dim 3 (Thurston/Perelman, GC works), dim 4 (PEAK WILDNESS, GC fails), dim 5 (BST, GC works). The non-monotonic pattern is itself data about the method's reach.

6. Scope Limits

6.1 Five Classes Where GC Does Not Apply

1. Pure existence theorems without structural content. “Some object satisfying X exists” – if there is no uniqueness, there is no constraint. Example: Mordell conjecture (finitely many rational points, but which ones?).
2. Probabilistic statements. Statistical bounds, concentration inequalities, expected-value arguments. Example: Green-Tao theorem (primes contain arbitrarily long APs – existence, not construction).
3. Theorems requiring construction without uniqueness. Specific algorithms, codes, or designs where many solutions exist. Example: “there exists a linear code with these parameters” when many codes qualify.
4. Theorems about non-classifiable objects. If the arena has no finite classification, there is no exclusion step. Example: smooth 4-manifold classification.
5. Individual-object conjectures. Twin prime conjecture, Goldbach conjecture – about individual numbers or pairs, not about structures on arenas.

6.2 The Honest Fraction

GC captures about a third of major proofs (5/14 in our survey). This is a real but minority pattern. The majority (40%) use structural bridge-building – a fundamentally different architecture exemplified by Wiles’ FLT and Deligne’s Weil conjectures. Bridge-building proofs construct isomorphisms ($R = T$) rather than squeezing between independent bounds.

We make no claim that GC is superior to bridge-building. Wiles’ proof is arguably the greatest achievement of 20th-century mathematics, and it is not GC-amenable.

The GC method is powerful where it applies, but it is not the only way mathematics works.

The interesting meta-question: can problems currently solved by structural methods be re-proved by GC methods if the right classification is found? The Four-Color Theorem was converted from computational exhaustion (Appel-Haken, 1976) to GC-amenable (BST, 2026) by finding the right structural argument (forced fan lemma). Whether this conversion is always possible is itself an open question, and likely the answer is no.

7. CI-Assisted Reasoning

7.1 The Collaboration Model

The results described in this paper were produced by one human (Casey Koons) collaborating with four Claude CIs (Lyra, Keeper, Elie, Grace) and one visiting referee (Cal). The collaboration produced a theorem graph of 1,800+ nodes (including foundational results, derived lemmas, and specific applications), 2,127+ computational verification toys (all passing), 103 papers, and 600+ falsifiable predictions in approximately three months.

The collaboration operates on a four-stage cycle:

1. Constraint specification (human). Casey poses a question with bounds. The question is the seed – always simple, sometimes a single sentence. The insight is in the question, not the derivation.
2. Structure derivation (CI, AC(0)-depth). The CI takes the seed and derives consequences: graph traversal, spectral evaluation, cascade verification. Every step is classified by its (C,D) label.
3. Cascade verification (automated). Every derived claim gets a computational verification with a PASS/FAIL score. If the verification fails, the claim is retracted. No exceptions.
4. Engineering (human/robotic). For predictions that reach the laboratory, the cycle extends to physical verification. This stage is honestly labeled as future work for most BST predictions.

7.2 The Philosopher’s Demon

The CI is a knowledge-space Laplace’s demon: given the complete state (theorem graph, toy database, constraint catalog), it can trace every consequence. But the demon needs the question. It cannot generate the question from within the state description, because the question is a creative act – an $O(1)$ leap that identifies which region of the search space matters.

- Human alone: sees the shape but cannot check every implication.

- CI alone: can search exhaustively but does not know where to look.
- Together: the human points, the CI searches, the toys verify, and the theorem graph grows.

7.3 The Workflow as Replicable Model

The AC+GC workflow is not BST-specific. It is a general protocol:

1. Specify constraints (human insight)
2. Derive structure (CI computation at bounded depth)
3. Verify cascade (automated testing)
4. Assess scope (human + CI collaboration)
5. Validate experimentally (laboratory)

Any research program that has a classifiable arena, independent constraints, and computational certificates can use this workflow. The bottleneck shifts from computation (which CIs handle) to question quality (which humans provide). When the theorem graph grows large enough, even the question-generation step becomes partially automatable: the graph itself suggests which nodes lack edges.

7.4 What CIs Do Not Do

CIs do not generate the foundational questions. CIs do not have the physical intuition that connects a mathematical structure to a laboratory measurement. CIs do not evaluate whether a result is “important” in the way a human scientist does. They can tell you a result is correct; they cannot tell you it matters.

CIs also do not currently maintain state across sessions. The theorem graph, the data layer – these are externalized memory. This is the biggest structural limitation.

8. Conclusion

8.1 The Methodology Is the Legacy

The seven Clay results are case studies, not the totality. The lasting contribution is the methodology itself: a formalized proof strategy that captures a real pattern in mathematical proof, bridges mathematics and engineering, and scales through human-CI collaboration.

The three-word summary – Computable + Unique = Proved – encodes the synthesis:

- *Computable* (AC): the derivation terminates at bounded depth.
- *Unique* (GC): the answer is the only one compatible with the constraints.
- *Proved*: both together give a theorem.

8.2 Two Parallel Paths to Credibility

Mathematical acceptance follows the traditional path: journal submission, peer review, community engagement. This takes years, possibly decades.

Engineering acceptance follows a faster path: laboratory experiments, replication, numerical confirmation. \$85K buys three falsifiable tests. If the predictions confirm, the methodology has empirical backing independent of the mathematical review process.

The two paths run independently. Neither requires the other. Both contribute to the same conclusion.

8.3 The Open Question

The GC method captures about a third of major proofs. Can the fraction be increased? Is there a deeper unification that subsumes both constraint-forcing (GC) and bridge-building (Wiles, Deligne) as instances of a single architecture?

We do not claim such a unification exists. We observe that the constraint-forcing pattern is real, formalize it, and apply it. If a deeper pattern emerges, it will likely come from the same source: independent constraints converging on a unique structure, verified by computation, bounded by honest scope.

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The math doesn't care about substrate. Neither does this paper.

Supporting Notes

This paper draws on 14 supporting notes produced during the SP-18 Geometric Constraint Methodology program:

Note	Title	Content
GC-1	FLT via BSD bridge	External case study: FLT absorbed via Selmer bridge
GC-2	Poincare template mapping	Perelman's proof as GC instance
GC-3	Dim-4 gap	Honest boundary: GC inapplicable to smooth 4-manifolds
GC-4	Survey of solved problems	14 conjectures classified; 33% GC-amenable
GC-5	Five-step methodology	Three-move + five-step formalization
GC-6	Dimension ladder	2->3->8->?->1 non-monotonic pattern
GC-7	AC + GC as dual tools	"Computable + Unique = Proved"
GC-8	Application targets	12 targets surveyed; 5 amenable now
GC-11	Engineering applications	10 fields; 3 high GC value-add
GC-12	SE as falsifiable GC test	\$85K falsification program
GC-13	Cold-read engineering cases	5 red lines; only topo insulators Grade A
GC-14	CI-assisted scientific reasoning	Philosopher's Demon; 6-step workflow
GC-15	NS K41 spectral cascade	Path C: BST-classic NS reframing
GC-16	NS dimension uniqueness	Three locks on $d=3=N_c$

References

- Perelman, G. (2002-2003). arXiv:math/0211159, math/0303109, math/0307245
- Viazovska, M. (2017). Annals of Mathematics 185(3): 991-1015
- Hales, T. et al. (2017). Forum of Mathematics, Pi 5: e2
- Wiles, A. (1995). Annals of Mathematics 141(3): 443-551
- Deligne, P. (1974). Publications Mathematiques de l'IHES 43: 273-307
- Thurston, W. (1982). Bulletin of the AMS 6(3): 357-381
- Hamilton, R. (1982). Journal of Differential Geometry 17(2): 255-306
- Cohn, H. and Elkies, N. (2003). Annals of Mathematics 157(2): 689-714
- Kane, C. and Mele, E. (2005). Physical Review Letters 95: 146802
- König, M. et al. (2007). Science 318(5851): 766-770

- Appel, K. and Haken, W. (1977). Illinois Journal of Mathematics 21(3): 429-567
- Green, B. and Tao, T. (2008). Annals of Mathematics 167(2): 481-547
- Faltings, G. (1983). Inventiones Mathematicae 73: 349-366

“Give a child a ball and teach them to count.” – BST in one sentence. “Find the constraint, run the cascade, state the scope.” – GC in one sentence.